

# Lab 09 – MATH 240 – Computational Statistics

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## Abstract

This lab explores the insightful question of why we find variance using  $n-1$  instead of  $n$ . We learn how to conduct a random sample of four different graphs and analyze the differences in the estimation of the variance using multiple sample sizes. We also determine the bias and the mean-squared error (MSE) to see which is truly the “best” method to calculate variance. Ultimately, using  $n-1$  to calculate the variance proves to be a better estimate.

**Keywords:** Variance, standard deviation, bias, mean-squared error, estimate

## 1 Introduction

Why do we calculate the variance using  $n-1$  and not  $n$ ? Oscar asked this question in the notes. It’s easy to recognize his confusion, as we are calculating an average — wouldn’t it make sense to divide by the sample size? However, there may be a reason that statisticians use  $n-1$  instead. The purpose of this lab is to graphically analyze if one formula is “better” for calculating the variance. We also aim to mathematically analyze our question by calculating the bias and the MSE, and seeing which formula produces a smaller value.

## 2 Methods

This lab uses a poisson, normal, binomial, and exponential distribution as avenues to discover which formula is better for determining the variance. We conduct random samples of each distribution using loops for sample sizes  $n = 5$ ,  $n = 10$ ,  $n = 50$ , and  $n = 100$ , and calculate the variance of the distribution using the  $n-1$  and  $n$  formulas in each loop. We stored these into data frames, and used the `dplyr` package (Wickham et al., 2022) to combine the data frames for each sample size into one data frame per distribution. To analyze the data graphically we use `ggplot()` in the `tidyverse` package (Wickham et al., 2019) to make box plots for each distribution that display the variance estimations by sample size and formula. We calculate the bias and the MSE separately, and add that to a table that numerically summarizes our results, and we use the `xtable` package (Dahl, 2022) to print this nicely.

## 2.1 Conducting Random Samples

The first thing we had to do for this lab was to create random samples of each known distribution (poisson, binomial, normal, exponential), and calculate the variance using the two formulas 1000 times for each sample size ( $n = 5$ ,  $n = 10$ ,  $n = 50$ ,  $n = 100$ ). We used a `for()` loop to do this. Before each loop, we saved the number of samples to be 1000, and created an empty tibble with 1000 rows that would store the values we calculate in the loop. Starting with a loop for a poisson distribution of sample size  $n = 5$ , we calculated a random sample with  $\lambda = 2$ , and stored the  $n-1$  variance formula results in the odd rows and the  $n$  variance formula results in the even rows. For each different sample size  $n$ , we conducted the same loop and stored it into another tibble, adjusting only the name of the tibble and the stored value for  $n$ . After getting four separate tibbles for each sample size from our loops, we used the `dplyr` package to bind the tibbles together to create one long tibble that encompassed all our data for the poisson distribution. This tibble had three columns — “estimate”, or the estimated variance, “type”, or the formula used to estimate the variance, and “n”, or the sample size used. This tibble was 4000 rows, 1000 rows for each loop.

To create the data for the other three distributions, we followed the same steps, creating 4 loops for each sample size that we stored in an individual tibble, and then combining those tibbles into one. The only things we adjusted were the parameters for each distribution (note that they were all specified so that the true variance would be 2) and the names of the tibbles.

## 2.2 Graphing our Results

After gathering our data into tibbles, we made box plots that displayed how well each formula estimated the variance by sample size. We used the `mutate()` function to convert the “n” column to a factor so that we could work with it as a categorical variable. Then we used `ggplot()` to make a box plot with `x` as the sample size, `y` as the estimated variance, and the `fill` as the formula used. We created this graph four times for each distribution, adjusting the data we used to the corresponding tibble we created.

## 2.3 Summarizing the Data into a Table

To summarize the data into a table, we took each distribution's data and grouped it by the sample size and the formula type. We reported the min, Q1, median, Q3, and max that we displayed in the box plots and calculated the bias by taking the mean of the estimates and subtracting it by 2 (the true variance). The sign would indicate if the data was overestimated or underestimated. We calculated the MSE by squaring the bias and subtracting the variance of our estimates. MSE is small when the bias and variance are small, which means the estimator is good. We created a table for each distribution to calculate those two values.

## 3 Results

Using the `patchwork` package, we combined the graphs regarding the variance estimators for each distribution. Each box plot is grouped by sample size, and the colors correspond to the respective formula. Each plot indicates very similar results, so I am adding one close-up example for simplicity. The four graphs together are included in the appendix.

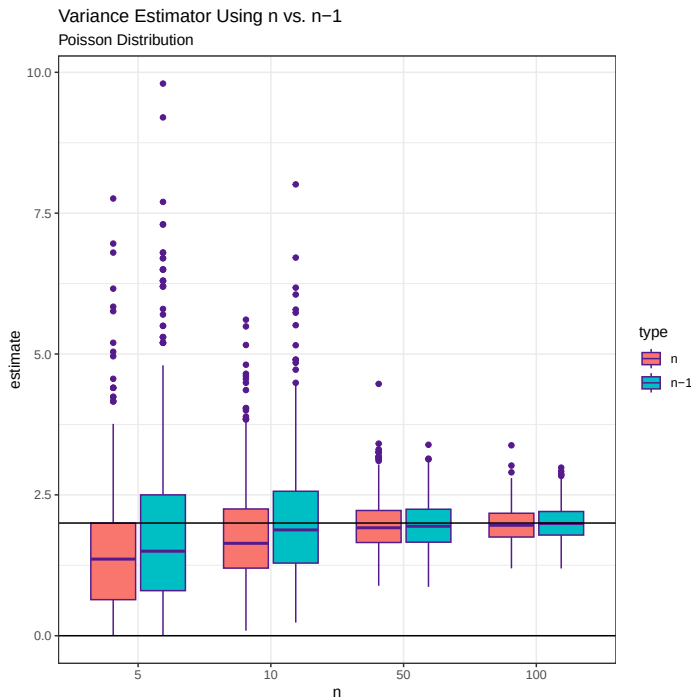


Figure 1: Close-up of poisson graph.

We can also use the `xtable` package to neatly print a table of our numerical summary and bias and MSE calculations for each distribution, grouped by sample size and type of formula. For simplicity, I will add an example of the poisson numerical summary with the calculated bias and MSE.

| n      | type | Min  | Q1   | Median | Q3   | Max  | Bias  | MSE  |
|--------|------|------|------|--------|------|------|-------|------|
| 5.00   | n    | 0.00 | 0.64 | 1.36   | 2.00 | 7.76 | -0.46 | 1.44 |
| 5.00   | n-1  | 0.00 | 0.80 | 1.50   | 2.50 | 9.80 | -0.05 | 2.46 |
| 10.00  | n    | 0.09 | 1.20 | 1.64   | 2.25 | 5.61 | -0.18 | 0.84 |
| 10.00  | n-1  | 0.23 | 1.29 | 1.88   | 2.56 | 8.01 | 0.03  | 1.10 |
| 50.00  | n    | 0.89 | 1.65 | 1.92   | 2.22 | 4.47 | -0.03 | 0.20 |
| 50.00  | n-1  | 0.87 | 1.66 | 1.94   | 2.24 | 3.39 | -0.03 | 0.19 |
| 100.00 | n    | 1.20 | 1.75 | 1.96   | 2.17 | 3.38 | -0.02 | 0.10 |
| 100.00 | n-1  | 1.19 | 1.79 | 1.99   | 2.20 | 2.98 | 0.02  | 0.11 |

## 4 Discussion

The results look very similar throughout all four distributions. When the sample sizes are small, it seems like both the `n-1` and `n` formulas underestimate the true variance of the data. However, one could argue that when the sample sizes are small, the `n-1` formula seems to be closer to the true variance than the `n` formula. Once the sample size increases, the variation between the estimates decreases drastically, and both formulas seem to estimate the true variance quite well.

It seems that the `n-1` formula for the variance is used because it tends to be a better estimator when the sample size is small, even though both formulas estimate the variance well when the sample size is large. This one difference, however, can mean a lot to a statistician, and in this situation, can be the deciding factor on why we use `n-1` rather than `n` to calculate the variance.

## References

- Dahl, D. B. (2022). *xtable: Export Tables to LaTeX or HTML*. R package version 1.8-4.
- Wickham, H., Averick, M., Bryan, J., Chang, W., McGowan, L. D., François, R., Grolemond, G., Hayes, A., Henry, L., Hester, J., Kuhn, M., Pedersen, T. L., Miller, E., Bache, S. M., Müller, K., Ooms, J., Robinson, D., Seidel, D. P., Spinu, V., Takahashi, K., Vaughan, D., Wilke, C., Woo, K., and Yutani, H. (2019). Welcome to the tidyverse. *Journal of Open Source Software*, 4(43):1686.
- Wickham, H., François, R., Henry, L., and Müller, K. (2022). *dplyr: A Grammar of Data Manipulation*. R package version 1.0.8.

## 5 Appendix

### Variance Estimator Using $n$ vs. $n-1$

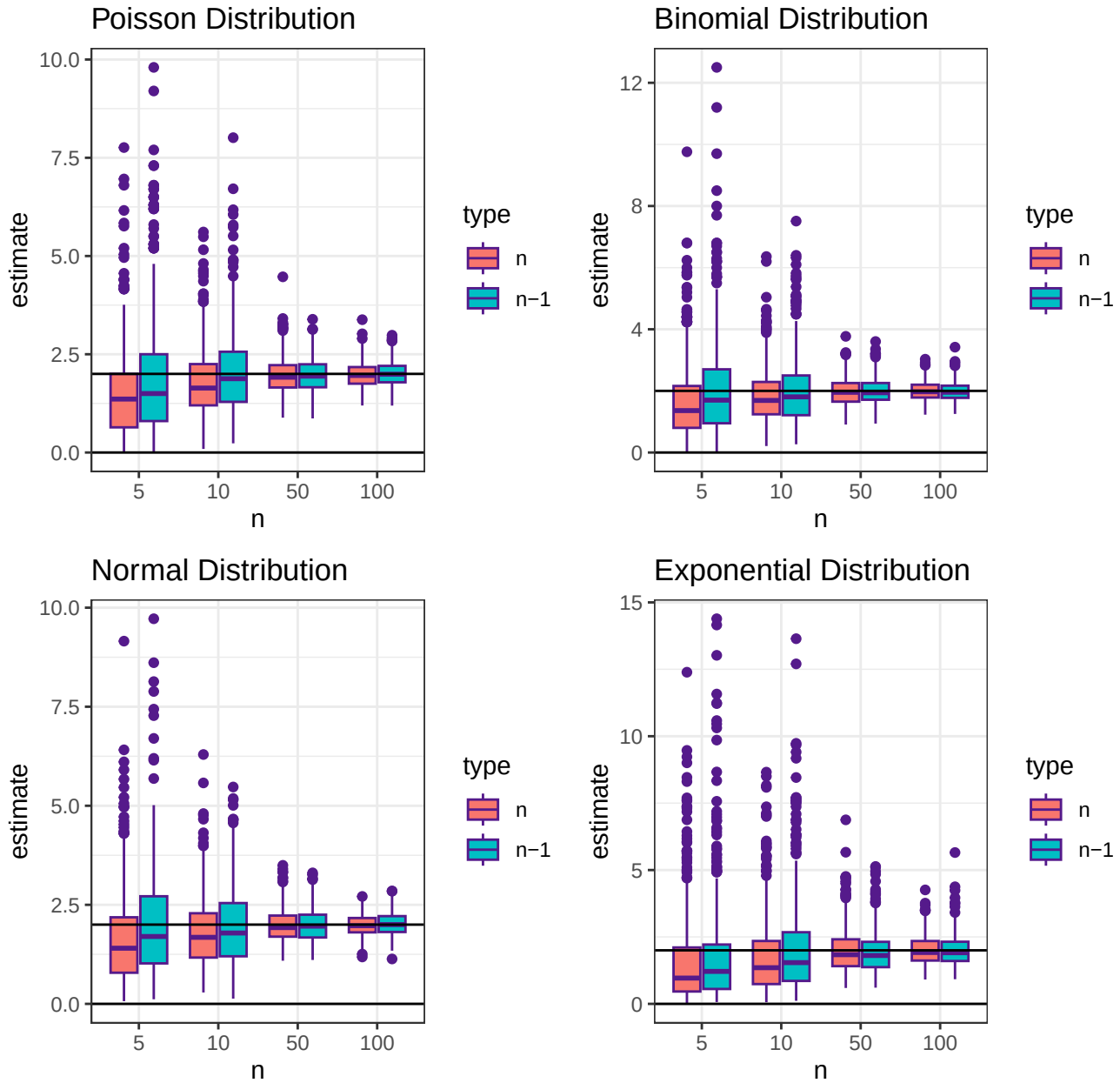


Figure 2: Poisson, binomial, normal, and exponential box plots together.